UNIFORM AXIS-PARALLEL STRIP HYPERGRAPHS WITH NO2-SHALLOW HITTING SET

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Editorial note. The original two PNG figures are included with their basenames. Their black lines are visible on the white PDF page; the transparent images were also checked in the official PDF page13 for the local cluster sizes.

Editorial note. Source542 puts conv around projected point sets inside a cardinality equal to a. The surrounding sentence535 and the original figure clearly specify a of the2a individual X_i,1 points on each side of the anchor; cardinality of a real interval is not intended. Original wording is retained with this clarification.

Editorial note. The construction may choose distinct x- and y-coordinates within every cluster. This preserves all strict cluster-order inequalities and lets a strip with fewer than m captured points be enlarged to exactly m, one point at a time. This is the extension step already invoked at592 and needed at545.

Editorial note. The source figures give |X|=2a, |B|=|C|=m-2a, |D|=2a-1, |E|=|F|=m-2a-b+1=a-b/2, |G|=|H|=b, |I|=m-2b, and |J|=|K|=2b. These are transcribed here for legibility; the original figures and prose remain the author proof.

Editorial note. The auxiliary clusters for distinct local gadgets occupy separate horizontal bands outside all global J_i strips, and the complete local gadget stays inside its own reserved vertical I interval, as required by source539/547.

Editorial scope. One geometric nonexistence construction for2-shallow hitting sets in m-uniform axis-parallel-strip hypergraphs for sufficiently large m. The complete local and global construction is counted, including both source figures. No3-shallow upper bound, polychromatic theorem or arithmetic-progression result is counted.

Allowed standard prerequisites (editorial).

- Finite planar range hypergraphs and general position. An axis-parallel strip captures exactly the input points between two vertical or two horizontal lines. The finitely many freely placed points can have distinct x- and y-coordinates while satisfying the author strict inequalities.. Choose strip boundaries between coordinates, and enlarge a captured set of at most m points to exactly m points by moving a boundary when needed.
- Elementary finite counting and set intersections. Track the exact cluster sizes and the consequences of every captured m-edge having one or two selected vertices.

Definition 1. *Denote by $H_{\geq m}$ or simply H_m the hypergraph obtained from H by deleting all hyperedges of size smaller than m, and denote by $H_{=m}$ the hypergraph consisting of the hyperedges of H with size exactly m. Similarly, for a hypergraph family $\mathcal{H}$ let $\mathcal{H}_m = \mathcal{H}_{\geq m} = \{H_{\geq m} : H \in \mathcal{H}\}$ and $\mathcal{H}_{=m} = \{H_{=m} : H \in \mathcal{H}\}$.*

Definition 2. *Let $H = (V, E)$ be a hypergraph, a $U \subset V$ vertex set $U \subset V$ is a c-shallow hitting set for a positive integer c if for every hyperedge $e \in E$: $1 \leq |e \cap U| \leq c$.*

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Definition 3. *Denote by $\mathcal{A}$ the hypergraph family which consists of those hypergraphs $H = (V, E)$ for which $V \subset \mathbb{R}^2$ is finite, and the edges are induced by axis-parallel strips: every $e \in E$ edge can be presented in the form $V \cap \{(x, y) \in \mathbb{R}^2 : x_0 < x < x_1\}$ or $V \cap \{(x, y) \in \mathbb{R}^2 : y_0 < y < y_1\}$ for some $x_0, x_1, y_0, y_1 \in \mathbb{R}$. Denote by $\mathcal{A}^*$ its dual: the family of such hypergraphs that the vertices are axis-parallel strips, and the edges are sets consisting of the strips containing a given point (x, y).*

Theorem 1. *For sufficiently large m there is an element in $\mathcal{A}_{=m}$ with no 2-shallow hitting set.*

Proof of Theorem 1. We construct a hypergraph in $\mathcal{A}$ with no 2-shallow hitting sets. The main arrangement of some vertices can be seen on Figure 1.

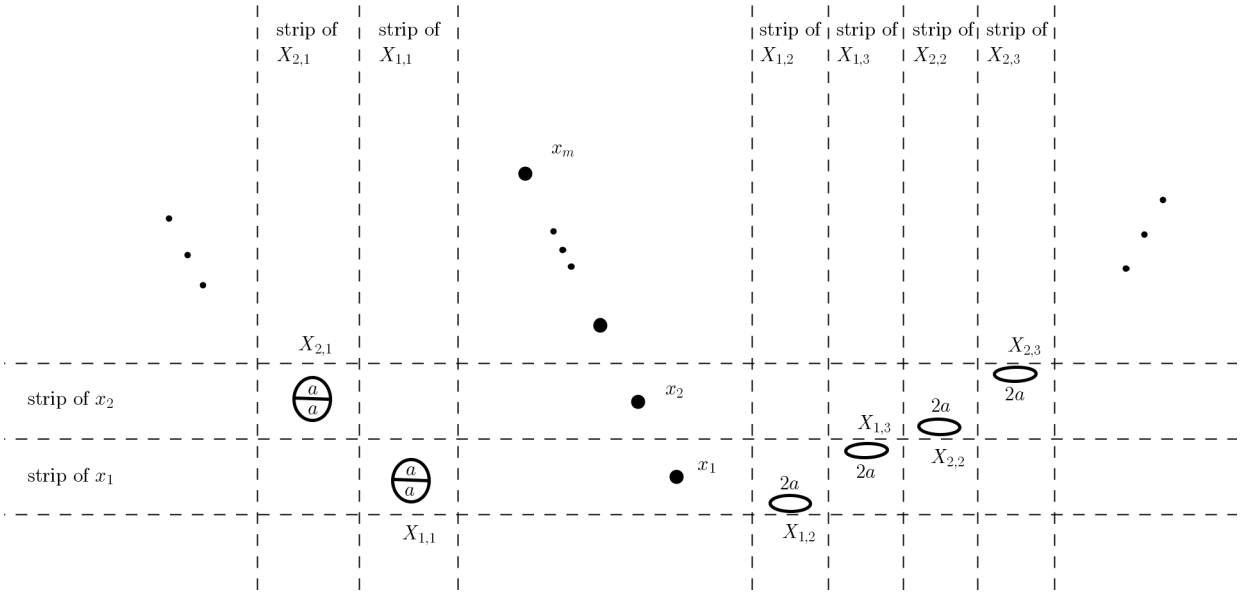


FIGURE 1. The construction for Theorem 1.

Let m be large enough, let $a = \lceil \frac{m}{3} \rceil - 1$, and b be 8 for $m \equiv 0 \mod 3$, 4 for $m \equiv 1 \mod 3$, and 6 for $m \equiv 2 \mod 3$, this satisfies $m = 3a + \frac{b}{2} - 1$. Later we will need that $m \geq 5b$.

First we take $X_0 = \{x_1, \ldots, x_m\}$, a set of m points on a southeast to northwest diagonal, as in the middle of Figure 1. We will leave empty a vertical strip containing them. Then we assign disjoint horizontal strips containing each x_i, and in all of these horizontal strips, we take 3 sets of points $X_{i,1}, X_{i,2}, X_{i,3}$ of size $2a$ as shown on Figure 1, reserving disjoint vertical strips containing each of these sets, and with all points from $X_{i,2}$ and a points from $X_{i,1}$ having smaller y-coordinates than x_i, and all points from $X_{i,3}$ and the other a points from $X_{i,1}$ having larger y-coordinates than x_i.

More precisely, the following properties are needed:

Take $3m + 1$ pairwise disjoint closed intervals $I_1, I_2, \ldots, I_{3m+1}$ on the x-axis, these correspond to the strips of X_0 and $\{X_{i,j}\}_{1 \leq i \leq m, 1 \leq j \leq 3}$. Take m pairwise disjoint intervals on the y-axis, $J_1, \ldots, J_m$, these correspond to the strips of $x_1, \ldots, x_m$. Let X_0 be a set of m points such that $\mathrm{conv}(\mathrm{proj}_1(X_0)) \subseteq I_{3m+1}$ and $\mathrm{proj}_2(x_i) \in \mathrm{int}(\mathrm{J_i})\,(1 \leq i \leq m)$. Let $\{X_{i,j}\}_{1 \leq i \leq m, 1 \leq j \leq 3}$ be sets of $2a$ points each such that $\mathrm{conv}(\mathrm{proj}_1(X_{i,j})) \subseteq I_{(i-1)\cdot 3+j}$ and

$\mathrm{conv}(\mathrm{proj}_2(X_{i,j})) \subseteq J_i \cap (\mathrm{proj}_2(x_i), \infty)$ if $j = 3$,
$\mathrm{conv}(\mathrm{proj}_2(X_{i,j})) \subseteq J_i \cap (-\infty, \mathrm{proj}_2(x_i))$ if $j = 2$,
$|\mathrm{conv}(\mathrm{proj}_2(X_{i,j})) \cap J_i \cap (\mathrm{proj}_2(x_i), \infty)| = |\mathrm{conv}(\mathrm{proj}_2(X_{i,j})) \cap J_i \cap (-\infty, \mathrm{proj}_2(x_i))| = a$ if $j = 1$.

It is enough to show that we can place additional points in the vertical strips in such a way that it forces the original set of $2a$ points to contain a point from a 2-shallow hitting set. Indeed, if this is the case, there is an element in X_0 from the hitting set, say x_i, and $X_{i,1}, X_{i,2}$, and $X_{i,3}$ each contain a point from the hitting set. Then there is a horizontal strip with $3a + 1 \leq m$ vertices which contains at least 3 points from the hitting set, namely, $\mathbb{R} \times [\mathrm{proj}_2(x_i), \max(J_i)]$ or $\mathbb{R} \times [\min(J_i), \mathrm{proj}_2(x_i)]$, depending on which half of $X_{i,1}$ contains a point from the hitting set, a contradiction.

For each $X \in \{X_{i,j} : 1 \leq i \leq m,\ 1 \leq j \leq 3\}$, the construction of this arrangement in a vertical strip is shown on Figure 2. These points are once again placed in separate horizontal strips for each $X_{i,j}$ (that is, the convex hulls of their projection to the y axis are pairwise disjoint closed intervals, disjoint even with $J_1, \ldots, J_m$). Fix $X \in \{X_{i,j} : 1 \leq i \leq m,\ 1 \leq j \leq 3\}$, we show that the needed arrangement indeed exists. Take 10 sets of points, $B, C, D, \ldots, K$, with cardinalities shown on Figure 2, so that for any elements $(x^B, y^B) \in B, (x^C, y^C) \in C, \ldots, (x^K, y^K) \in K$ and $(x^X, y^X) \in X$ we have

$$x^I < x^K < x^J < x^G < x^E < x^D < x^H < x^F < x^C < x^X < x^B,$$
$$y^X < y^K < y^J < y^I < y^G < y^H < y^E < y^B < y^D < y^C,$$
$$y^H < y^F < y^B.$$

Therefore, if we take neighboring strips on Figure 2, then the union of the contained sets is captured by an axis-parallel strip.

Moreover, we arrange the points of $B = \{b_1, b_2, \ldots, b_{m-2a}\}$ and $C = \{c_1, c_2, \ldots, c_{m-2a}\}$ on a diagonal line such that for $b_k = (x^B_k, y^B_k)$ and $c_l = (x^C_l, y^C_l)$ $(1 \leq k, l \leq m-2a)$ we have

$$x^C_{m-2a} < x^C_{m-2a-1} < \ldots < x^C_1 < x^B_{m-2a} < x^B_{m-2a-1} < \ldots < x^B_1,$$
$$y^B_1 < y^B_2 < \ldots < y^B_{m-2a} < y^C_1 < y^C_2 < \ldots < y^C_{m-2a}.$$

This implies that if we take $B' = \{b_k, b_{k+1}, \ldots, b_{m-2a}\}$, the last few elements of B, and $C' = \{c_1, c_2, \ldots, c_l\}$, the first few elements of C, then both $B' \cup C' \cup D$ and $B' \cup C' \cup X$ are captured by an axis-parallel strip.

Now let S be a 2-shallow hitting set, and suppose indirectly that X does not contain an element from S. Then B contains a point from the hitting set, because $B \cup X$ is captured by a vertical strip of size m. Take the largest k such that $b_k \in S$, and take $B' = \{b_{k+1}, \ldots, b_{m-2a}\}$, and $C' = \{c_1, c_2, \ldots, c_k\}$. Then $B' \cup C' \cup X$ is captured by a vertical strip, $|B' \cup C' \cup X| = m$, and $S \cap (B' \cup X) = \emptyset$, so $S \cap C' \neq \emptyset$. Therefore, $|S \cap (\{b_k\} \cup B' \cup C')| \geq 2$, which implies that $D \cap S = \emptyset$, because $\{b_k\} \cup B' \cup C' \cup D$ is captured by a horizontal strip.

Observe that $|D \cup E \cup G| = |D \cup E \cup H| = |D \cup F \cup H| = m$, and that these sets are all captured by an axis-parallel strip, thus $E \cup G$, $E \cup H$ and $F \cup H$ each contain a point from S. Notice that $a - \frac{b}{2} = m - 2a - b + 1 = |E| = |F|$, so $|E \cup F \cup H| = 2a$, which implies $\big|(E \cup F \cup H) \cap S\big| \leq 1$, because otherwise $E \cup F \cup H \cup B$ would form a hyperedge of cardinality m containing at least 3 points from the hitting set.

This implies $E \cap S = \emptyset$, hence G and H each contain a point from S, so $I \cap S = \emptyset$. Thus J and K each contain a point from S, so $|S \cap (G \cup J \cup K)| \geq 3$. Assuming $m \geq 5b$, $G \cup J \cup K$ can be extended to a hyperedge of cardinality m, which is a contradiction. □

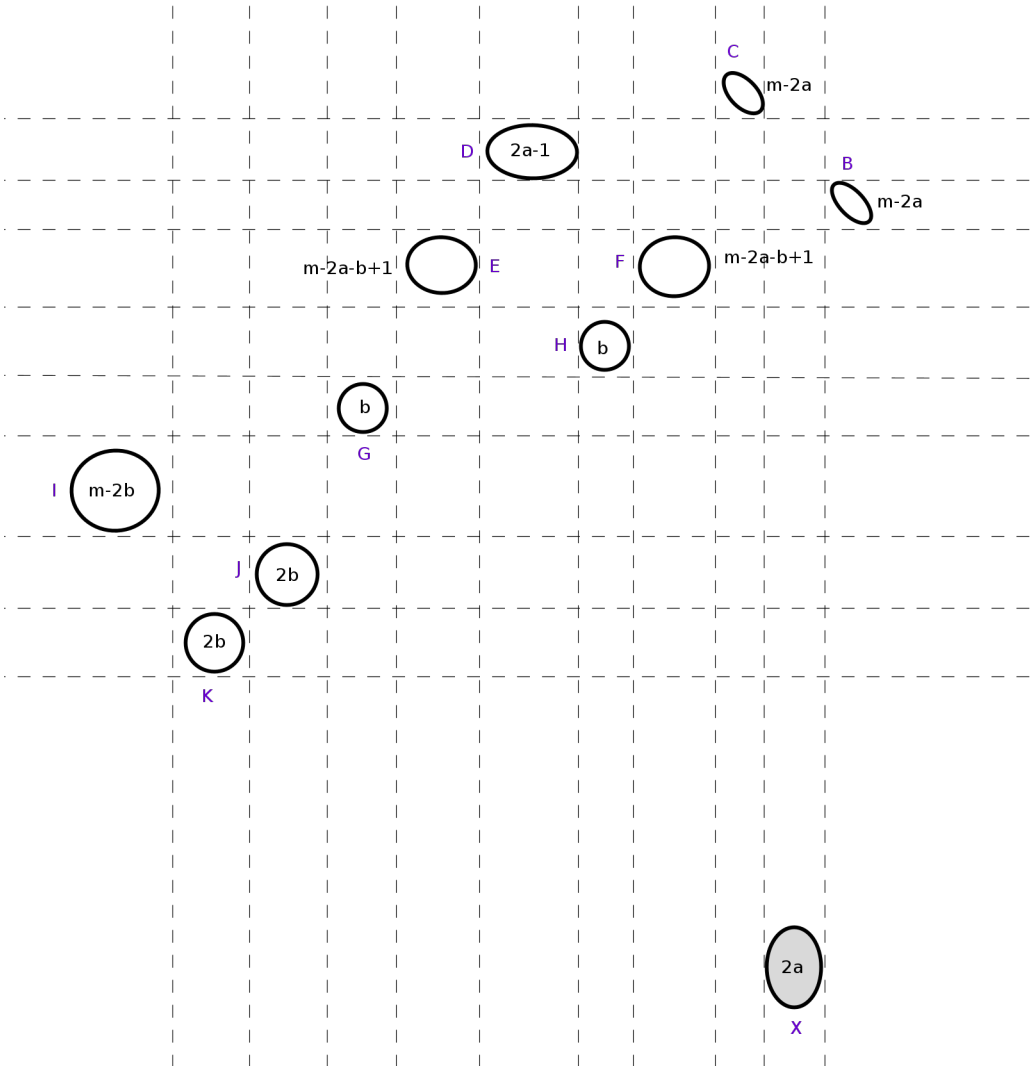


FIGURE 2. The vertical strip of a $2a$-set X